

Annex 93C

(normative)

Receiver interference tolerance

This annex defines a test setup (see 93C.1) and method (see 93C.2) for testing receiver interference tolerance. The PMD clause that invokes this method specifies the following items:

- a) Constraint limit values for peak-to-peak voltage, the pre-cursor peaking ratio, and the post-cursor peaking ratio for test setup,
- b) Lower frequency bound for the noise spectral density constraints (f_{NSDI}),
- c) Jitter parameters to be measured in test method step 3,
- d) Target Channel Operating Margin (COM) and RSS_DFE4 values for the test system in test method step 7,
- e) COM parameter table in test method step 7,
- f) Jitter transformation method in test method step 7,
- g) Test pattern in test method step 9, and
- h) Test system frequency response.

NOTE—The intent of the interference tolerance test is to ensure that the PHY receiver operates correctly with transmitter parameters anywhere within the specified limits including the case where all parameters are at the specified limits. Testing of the receiver with transmitter parameters beyond the specified limits may be helpful to determine margin or to provide comparative metrics, but failure of the receiver to operate correctly under these conditions is not to be interpreted as non-compliance.

93C.1 Test setup

The interference tolerance test is performed with the setup shown in Figure 93C–2 or its equivalent. Calibration and characterization of the various elements in the test setup is accomplished using the test configurations in Figure 93C–3, Figure 93C–4, Figure 93C–5, and Figure 93C–6.

The transmitter is functionally and parametrically compliant to the requirements of the invoking PMD clause. The ISI channel emulates the frequency dependent loss of a backplane channel. The channel noise source emulates crosstalk, transmitter noise, and unequalizable signal distortions introduced by a channel.

The transmitter output, as measured at TP0a, meets all transmitter specifications as indicated by the invoking PMD clause. In addition, the transmitter output, as measured at TP0a, is constrained such that for any transmitter equalizer setting the maximum differential peak-to-peak voltage, the pre-cursor peaking ratio, and the post-cursor peaking ratio are constrained as indicated by the PMD clause that invokes this method.

The channel noise source has an adjustable output such that the level may be set according to the test procedure. The noise produced by the channel noise source is measured directly at the output of the noise source (see Figure 93C–6). The noise is Gaussian with a crest factor of at least 5. The noise spectral density, $NSD(f)$, is normalized and constrained according to the relations in Equation (93C–1), where f_b is the symbol rate and f_{NSDI} is specified by the PMD clause that invokes this method. $NSD(f)$ is in units of V^2/Hz . The average noise spectral density, $NSD_{average}$, is determined according Equation (93C–2). An example constraint template with f_{NSDI} equal to $0.08f_b$ is illustrated in Figure 93C–1.

$$\left. \begin{aligned} 10\log_{10}\left(\frac{NSD(f)}{NSD_{average}}\right) &< 3 \\ 10\log_{10}\left(\frac{NSD(f)}{NSD_{average}}\right) &> -3(1 - 1.2f/f_b) \end{aligned} \right\} f_{NSD1} \leq f \leq f_b/2 \quad (93C-1)$$

$$NSD_{average} = \frac{\int_{f_{NSD1}}^{f_b/2} NSD(f) df}{f_b/2 - f_{NSD1}} \quad (93C-2)$$

The receiver on one lane at a time is tested for compliance. The input to the receiver on each of the other lanes is generated by a transmitter with similar levels and equalization settings and transmitted through a similar channel, such that the input signals are similar to the input signal on the lane under test.

NOTE—FEXT and NEXT in the test setup are not accounted for in the test channel calibration (see 93A.2). It is recommended that the test setup be designed to minimize these effects.

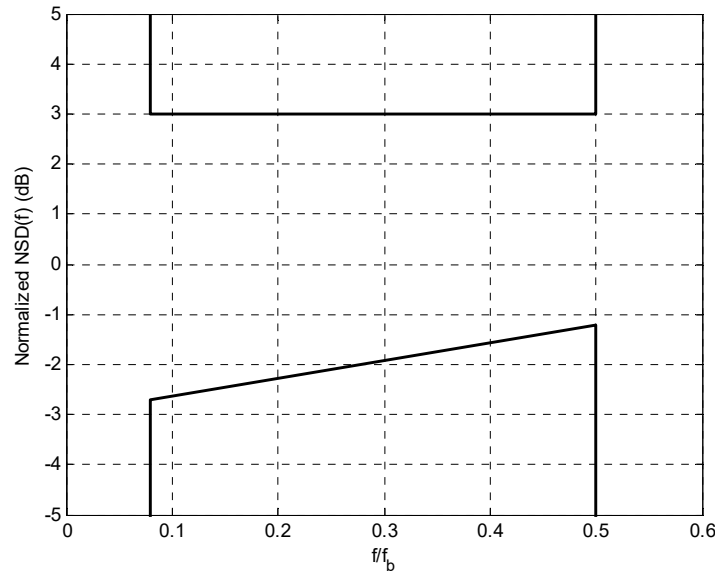


Figure 93C-1—Example NSD(f) constraint template

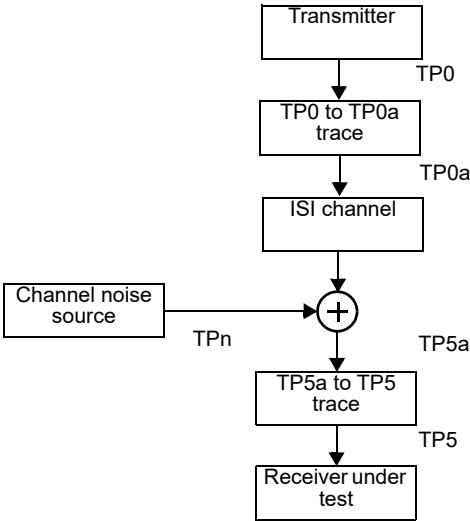


Figure 93C-2—Interference tolerance test setup

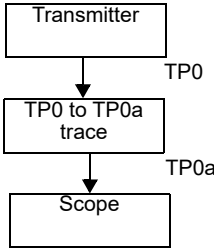


Figure 93C-3—Interference tolerance transmitter test setup

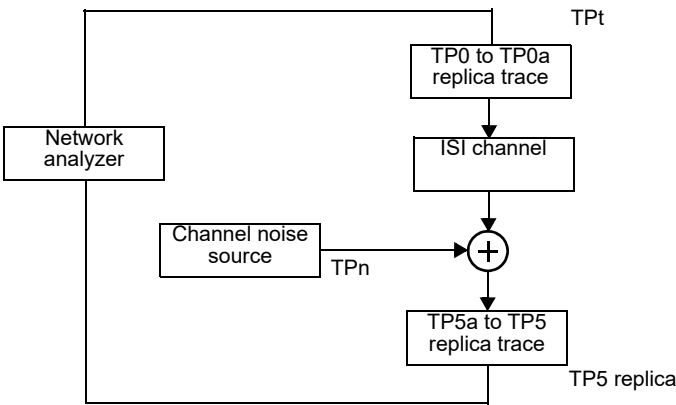


Figure 93C-4—Interference tolerance channel s-parameter test setup

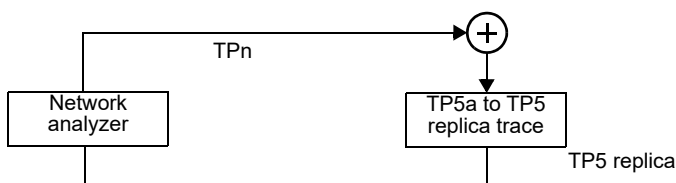


Figure 93C-5—Interference tolerance channel noise path test setup

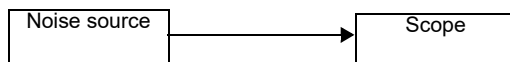


Figure 93C-6—Interference tolerance channel noise level test setup

93C.2 Test method

The interference tolerance test is performed using the following method:

- 1) Set the channel noise source to zero.
- 2) Using the test setup in Figure 93C-2, initiate the training sequence, allow the training sequence to complete, and retain the resulting transmitter tap coefficients.
- 3) Measure the jitter parameters relevant to the PMD clause that invokes this method that are to be used to set the value of σ_{RJ} and A_{DD} in step 7.
- 4) Measure the noise parameters relevant to the PMD clause that invokes this method that are to be used to set the value SNR_{TX} .
- 5) Using the test setup in Figure 93C-4 (also see Figure 93A-2), measure the scattering parameters, $S^{(tc)}$, of the test channel (TPt to TP5 replica).
- 6) Using the test setup in Figure 93C-5 (also see Figure 93A-2), measure the scattering parameters, $S^{(nc)}$, of the noise addition network (TPn to TP5 replica).
- 7) Using the procedure defined in 93A.2: (a) determine the receiver noise level, σ_{bn} , required to achieve the COM value specified in the PMD clause that invokes this method, and (b) verify that RSS_DFE4 is greater than or equal to the value specified in the PMD clause that invokes this method. The procedure is based on the calculation of COM, which uses the parameters defined in the COM parameter table in the PMD clause that invokes this method with the following exceptions. The value of σ_{RJ} and A_{DD} are set based on a transformation of measured parameters as specified in the PMD clause that invokes this method. The value of SNR_{TX} is set based on a transformation of the measured parameters specified in the PMD clause that invokes this method. In the COM computation the transmitter package model is included only if a compliant transmitter with a similar termination is used. If a transmitter with high quality termination is used, in the COM calculation, the termination is modeled as ideal and a Gaussian low pass filter is added to Equation (93A-19), which has the same 20% to 80% transition time as the transmitter measured at TP0a.
- 8) Using the test setup in Figure 93C-6, measure the channel noise voltage σ_{bnm} and adjust it so that it equals σ_{bn} determined in step 7. The channel noise voltage is determined from the measured $NSD(f)$ according to Equation (93C-3).
- 9) Using the test setup in Figure 93C-2, the transmitter taps as determined in step 2, and the channel noise as determined in step 7, configure the transmitter to transmit the test pattern specified in the PMD clause that invokes this method. Also configure the transmitters of the PMD under test to transmit the same test pattern, with their transmitters in the preset condition.